

MIP_AGENT_ARCHITECTURE.md

Mainframe Intelligence Platform (MIP)

Multi-Agent Architecture using Google ADK

Version: 1.0

Status: Core Platform Architecture

Classification: Strategic Implementation Document

Owner: MIP Engineering Team

1. Executive Summary

MIP is implemented as a Multi-Agent System.

Instead of building one massive application, MIP decomposes responsibilities into specialized agents.

Each agent owns a specific responsibility.

Benefits:

- Scalability
 - Explainability
 - Independent evolution
 - Easier testing
 - Better AI integration
 - Future autonomous modernization
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2. Agent Philosophy

Traditional Architecture:

Monolithic Application

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Everything Everywhere

MIP Architecture:

Specialized Agents

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Coordinated Intelligence

↓

Modernization Platform

3. High-Level Agent Architecture

Repository

↓

Discovery Agent

↓

Parser Agents

↓

Metadata Agent

↓

Graph Agent

↓

Lineage Agent

↓

Impact Analysis Agent

↓

Reasoning Agent

↓

Modernization Architect Agent

4. Core Agent Categories

MIP consists of five layers.

```
Layer 1
Discovery

Layer 2
Parsing

Layer 3
Knowledge Construction

Layer 4
Intelligence

Layer 5
Modernization
```

5. Discovery Layer

Responsible for repository understanding.

Agents:

```
DiscoveryAgent

RepositoryScannerAgent

FileClassifierAgent
```

DiscoveryAgent

Purpose:

Discover repository assets.

Inputs:

```
Repository Path
```

Outputs:

File Inventory
Directory Structure
Technology Inventory

Responsibilities:

- Repository scanning
- File identification
- Inventory creation

RepositoryScannerAgent

Purpose:

Perform deep repository scan.

Outputs:

Programs
Copybooks
JCL
Scripts
Configs
Documentation

FileClassifierAgent

Purpose:

Classify discovered files.

Examples:

COBOL

JCL

PL/I

RPG

Java

SQL

YAML

JSON

6. Parsing Layer

Responsible for source code understanding.

Agents:

CobolParserAgent

JCLParserAgent

CopybookParserAgent

SQLParserAgent

JavaParserAgent

DocumentationParserAgent

CobolParserAgent

Purpose:

Extract COBOL metadata.

Outputs:

Programs

Calls

Tables

Files

Variables

Business Rules

JCLParserAgent

Purpose:

Extract job metadata.

Outputs:

Jobs

Steps

Datasets

Program Execution Chains

CopybookParserAgent

Purpose:

Extract data definitions.

Outputs:

Structures

Fields

Layouts

Data Types

SQLParserAgent

Purpose:

Extract database metadata.

Outputs:

Tables

Columns

CRUD Operations

DocumentationParserAgent

Purpose:

Extract business context.

Inputs:

PDF

DOCX

TXT

Markdown

Outputs:

Business Rules

Capabilities

Domains

7. Knowledge Construction Layer

Responsible for metadata and graph creation.

Agents:

MetadataAgent

RelationshipAgent

GraphBuilderAgent

MetadataAgent

Purpose:

Normalize metadata.

Outputs:

Canonical Metadata Model

Stores:

SQLite

RelationshipAgent

Purpose:

Infer relationships.

Examples:

CALLS

READS

WRITES

EXECUTES

USES

GraphBuilderAgent

Purpose:

Build enterprise knowledge graph.

Technology:

NetworkX

Outputs:

Knowledge Graph

Graph Metrics

8. Intelligence Layer

Responsible for analysis and reasoning.

Agents:

LineageAgent

ImpactAnalysisAgent

ReasoningAgent

RiskAssessmentAgent

LineageAgent

Purpose:

Construct lineage paths.

Outputs:

Data Lineage

Execution Lineage

Business Lineage

ImpactAnalysisAgent

Purpose:

Calculate blast radius.

Outputs:

Impact Reports

Dependency Maps

Affected Assets

RiskAssessmentAgent

Purpose:

Assess modernization risk.

Outputs:

Risk Scores

Risk Factors

Mitigation Plans

ReasoningAgent

Purpose:

Generate insights.

Questions:

What is important?

What is risky?

What belongs together?

What should be modernized first?

Outputs:

Recommendations

Insights

Rankings

9. Modernization Layer

Responsible for future-state architecture.

Agents:

CapabilityDiscoveryAgent

ServiceDiscoveryAgent

MigrationPlanningAgent

ModernizationArchitectAgent

CapabilityDiscoveryAgent

Purpose:

Discover business capabilities.

Outputs:

Capabilities

Domains

Ownership Maps

ServiceDiscoveryAgent

Purpose:

Identify service boundaries.

Outputs:

Microservice Candidates

API Candidates

Event Candidates

MigrationPlanningAgent

Purpose:

Create modernization waves.

Outputs:

Wave 1

Wave 2

Wave 3

Wave 4

ModernizationArchitectAgent

Purpose:

Generate target architecture.

Outputs:

Reference Architecture

Migration Roadmap

Modernization Recommendations

10. Agent Communication

Pattern:

Agent

↓

Task

↓

Result

↓

Evidence

↓

Next Agent

Every result contains:

```
{
  "agent": "",
  "result": {},
  "confidence": 0.0,
  "evidence": []
}
```

11. Google ADK Mapping

Suggested Structure:

```
src/mip/agents/  
  
    discovery/  
  
    parsing/  
  
    metadata/  
  
    graph/  
  
    lineage/  
  
    impact/  
  
    reasoning/  
  
    modernization/
```

Base Agent Contract

```
class BaseAgent:  
  
    def run(self, context):  
        pass  
  
    def validate(self):  
        pass  
  
    def explain(self):  
        pass
```

12. Shared Context Model

All agents share:

```
class MIPContext:  
  
    repository
```

metadata

graph

lineage

impact

recommendations

13. Agent Orchestration Flow

Discovery

↓

Parsing

↓

Metadata

↓

Relationships

↓

Graph

↓

Lineage

↓

Impact

↓

Reasoning

↓

Modernization

14. Explainability Requirements

Every agent must provide:

Input

Output

Evidence

Confidence

Reasoning

No black-box decisions allowed.

15. Future Agent Roadmap

Phase 1

Rule-Based Agents

Phase 2

LLM-Assisted Agents

Phase 3

Collaborative Agents

Phase 4

Self-Improving Agents

Phase 5

Autonomous Modernization Agents

16. Enterprise Scale Targets

Support:

100,000+ Programs

20,000+ Jobs

500,000+ Tables

10M+ Relationships

17. Success Criteria

A user should be able to ask:

Analyze this repository and recommend a modernization strategy.

And MIP should automatically:

1. Discover Assets
2. Parse Source Code
3. Build Metadata
4. Create Knowledge Graph
5. Generate Lineage
6. Perform Impact Analysis
7. Assess Risk
8. Discover Services
9. Create Migration Waves
10. Produce Modernization Roadmap

with explainable evidence.

MIP Principle

Agents collect knowledge.

Graphs connect knowledge.

Reasoning creates intelligence.

Modernization creates business value.

End of Document